
A Hybrid Spatio-Temporal Pipeline for NYC Crash Hotspot Discovery and Severity Prediction

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Abstract

The data science workflow outlined in this paper addresses two issues in existing research and practice. First, road traffic accidents are a significant hazard in densely populated areas such as cities, but the vast majority of city governments do not provide open access to their accident databases; second, although most researchers use supervised learning techniques to predict crash severity; a few studies have employed or discussed unsupervised geospatial analysis to identify high-risk areas of the city. This paper presents a pipeline of data science processes applied to the New York City Motor Vehicle Collisions Dataset (2021-2022), which applies both unsupervised geospatial clustering and supervised classification of crash severity. The process begins with a single preprocessing stage that prepares the NYC MVC crash records into four feature categories: temporal/cyclical time encoding; weather coding; spatial coding; and a daily-weather join. After applying the Elbow and Silhouette diagnostics to the training set of crash locations to find $k = 4$, the researcher uses k-means to cluster crash locations into four groups and validates these results against a DBSCAN baseline based on the Haversine distance. Following validation, the researcher defines the cluster identity as a categorical variable and feeds it into three sequential logistic regression, random forest, and xgboost classifiers to determine whether the crash falls into the “serious” category. Each classifier was trained using methods designed to correct for the extremely large imbalance of classes found in the original data (2.75 percent “serious”). The researcher then evaluated each model based on performance metrics including Precision-Recall AUC (PR-AUC), Receiver Operating Characteristic Area Under Curve (ROC-AUC) scores, Balanced Accuracy, Calibration Statistics, SHAP feature attributions, and a Feature Group Ablation Test. Overall, there were three key results from this research: (1) the four identified K-means clusters cover approximately a 28% relative difference in serious-crash rates; (2) Class-Weighted Logistic Regression produced the highest balanced accuracy (0.585), suggesting that the choice of how to treat imbalanced classes is significantly more important than the capacity of the model being used; and (3) once raw latitude and longitude are available, gradient-boosted tree models can already extract the underlying spatial structure on their own, making the K-means cluster identity largely redundant for them as a feature.

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1 Introduction

Traffic accidents are among the highest frequency, most impactful public health events in cities. In addition to publishing hundreds of thousands of collision reports annually through NYC Open Data [12], the NYC Open Data platform provides collision information about incidents, which is detailed in both time and space. However, the data itself, while extensive in detail for location and time, is provided typically in a non-relational format (i.e., flat CSV file) without being combined with additional relevant signals such as weather or road geometry; without a structured management system for the data; and without defined baselines to connect unsupervised discovery of hotspots to supervised predictions of accident severity. Consequently, downstream analysis tends to be “ad hoc,” difficult to reproduce; and rarely includes suitable metrics that account for the extreme class imbalance inherent in predicting severe accidents.

This paper outlines an integrated data science workflow from raw feeds of NYC collision data and weather conditions to a reproducing analytical object well-suited to answering two complementary queries: **Q1**) *Where and at what times do severe accidents occur?* and **Q2**) *Can we use the available contextual attributes to determine if a particular incident will likely be classified as a severe accident?*

Our contributions are as follows:

- Our code and data are entirely reproducible via an easily reproduced parquet database (SQLite cache), deterministic seed values, as well as a one-line command to reproduce all figures, tables, and metrics presented within this paper.²
- Using KMeans clustering (we used Elbow & Silhouette diagnostic techniques to find our number of clusters), we performed unsupervised spatial hotspot detection on the joined crash-weather dataset and measured how the cluster identities corresponded with rates of serious crashes.
- We designed a supervised multi-layer classification model (majority-class baseline; logistic regression; random forest; xgboost; xgboost + SMOTE; xgboost w/ threshold optimization) based upon a chronological train-test split and evaluated its performance using PR-AUC, ROC-AUC, Balanced Accuracy, Calibration Metrics, and SHAP Feature Attribution analysis.
- Finally, we conducted an Ablation study to identify the incremental contribution of each of the four feature groups. The results show that time-based features are most valuable and that spatial features have very little additional benefit beyond what can be obtained by knowing (lat, lon).

2 Related Work

We group prior work into three categories that map directly onto the components of our pipeline: (i) spatio-temporal clustering of crashes, (ii) supervised crash-severity prediction, and (iii) hybrid frameworks combining clustering and classification with explainability.

Crash Hotspot Detection. Hotspot detection was one of the first use cases for spatial analysis in road safety. Anderson used K-means clustering along with Kernel Density Estimation to find high-risk locations based on accidents in London [2]. Prasannakumar extended their work into GIS-based spatiotemporal approach [14]. Recently, density-based approaches like DBSCAN [6] were utilized to determine local hotspot areas by considering the geometry of the data at hand. The authors found it beneficial to link crashes to environmental features (such as weather and POIs), which has already shown promise from previous studies (Moosavi [11], etc.). This study follows the same line of research; however, we propose combining both KMeans and Haversine distance DBSCAN in an integrated pipeline.

Crash Severity Prediction. The supervised severity classification has been approached by an assortment of different models. Abdelwahab et al. (2001) utilize artificial neural networks to estimate the injury severity for drivers [1]. Iranitalab et al. (2017), conducted a systematic comparison of a variety of models that include Multinomial Logit Model, Nearest Neighbor Model, Support Vector Machine, Random Forest Models using U.S. crash data [8]. Theofilatos et al. (2014), reviewed how

²Code: <https://github.com/Vishal-Nagamalla/Traffic-Hotspot-Final>

weather and traffic condition affect injury severity [15]. There are two common themes throughout the studies on the injury severity classification; first it is difficult to recover the severity signals from the limited number of features available post-crash as well as, secondly due to the large class imbalance that exist within most crash datasets.

Hybrid and Explainable Frameworks. A number of studies have begun to combine unsupervised segmentation with supervised prediction and apply tools for explaining the predictions (such as SHAP [9]), which was originally introduced as a general framework for interpreting machine-learning models and has since been used in many other areas including safety analytics. Techniques that are aware of imbalance (such as SMOTE [4]) and threshold tuning, when the positive class is rare, have become standard when working with imbalanced datasets [7]. Our research combines these concepts by using an unsupervised clustering method to extract spatial patterns from the data, then applying a tree ensemble classifier ladder [3, 5] to predict crash severity using both the original features and the cluster identity that was produced by the clustering step, and finally using SHAP to provide explanations for the top performing tree-based model.

3 Methodology

3.1 Datasets

We used two publicly available datasets. The NYC Motor Vehicle Collisions dataset [12], was converted into a CSV file with each row representing one motor vehicle collision, which included the date/time of the crash, Borough, Zip code, Latitude/Longitude of the location where the crash occurred, On/Off street name(s), Number of persons Injured or Killed. The NYC Hourly Weather [13] dataset includes data on hourly temperature, Precipitation, Rain, Cloud Cover and Wind Speed. However although the Crash Data spans from 2021-2025 and the weather data spans from 2016-2022, we utilized both datasets for the years 2021-2022 (a two year span) so that there would be a weather context associated with each crash. After filtering out bad time stamps, coordinates that were outside the bounds of the NYC geographic area, and after successfully joining the weather data with the crash data, we retained **177,313 crashes** and **2,490 weather days** of data. Of those crashes approximately 42k had an unknown value for the borough field; they will be modeled but will not be included when providing the results by borough shown in Table 1.

3.2 Data Preprocessing

We apply the same pipeline for cleaning data as we do to preprocess our training data. The entire pipeline is encapsulated in a single sklearn ColumnTransformer/Pipeline so that no transform will ever see more than the one training fold and therefore cannot leak any test information.

Cleaning. We exclude crash records whose timestamp could not be parsed, which have missing coordinates, or coordinates that are outside of New York City’s bounding box $(-74.3, -73.6) \times (40.4, 41.0)$, and then remove duplicates based on the unique collision id.

Severity Label. We define a binary label $y_i = 1\{\text{deaths}_i > 0 \vee \text{injuries}_i \geq 3\}$, matching the operational definition of a serious crash used by NYC’s Vision Zero program. The serious rate on the joined dataset is 0.0275.

Weather Aggregation. Because the source weather feed is hourly, we aggregate to a daily record with maximum, minimum, and mean temperature (in °F), summed precipitation (in mm), summed rain, mean cloud cover, and maximum wind speed. We also derive boolean buckets: `is_rainy` ($\text{precip} > 1 \text{ mm}$), `is_heavy_rain` ($\text{precip} > 10 \text{ mm}$), `is_freezing` ($T_{\min} \leq 32^\circ\text{F}$), `is_hot` ($T_{\max} \geq 85^\circ\text{F}$), `is_cloudy`, and `is_windy`.

Encoding cyclical time attributes. The date and time of every incident was analyzed by breaking it down into three main parts. Hour, Day of Week, Month were calculated for all incidents. Each part of this information was represented using two separate numbers that corresponded to their sine and cosine values respectively at different frequencies such that, the same frequency is used for 12 o’clock AM and 11 o’clock PM. In addition to the previously mentioned number encoding, an integer

representation of hour was kept in order to be able to create histograms. Flags representing if a collision occurred during a weekend, during rush hours or during nighttime were added to the data as well.

Adding Spatial Information. A spatial distance attribute called `dist_to_midtown_km` was created using the Euclidean distance formula to calculate distances from a given point. This distance measure represents how far away in kilometers an incident location is from Times Square. The coordinates for Times Square are (40.7580, -73.9855). Two additional attributes `lat_bin` and `lon_bin` hold a 0.01° grid identifier for latitude and longitude as a coarse spatial grouping signal.

Using One-Hot and Standardization Transforms. All boroughs have been one-hot encoded. Also in our second model (cluster-augmented), all cluster ids obtained via k-means will be one-hot encoded. As for standardizing the numerical data, median imputation will be applied to replace missing values, then the means of the numerical variables will be subtracted and divided by the standard deviations in order to obtain the desired distribution. These transformations will be fitted to the training set of the split over time.

Train/Test Split. In order to follow established guidelines within the field of spatiotemporal prediction, the data was split based on time. We used the first 80% of all crashes ordered by `crash_datetime` as our training set. The last 20% of all crashes were placed into our test set. The majority of the training set that is 90% was then used to fit the model. A remaining 10% of the training set was held out as an internal validation slice to allow us to find optimal thresholds.

3.3 Hotspot Discovery via KMeans and DBSCAN

We apply unsupervised clustering in (latitude, longitude) space, fitted on the training fold only.

Choosing k for KMeans. We sweep $k \in \{2, \dots, 12\}$ and report both the within-cluster sum-of-squares (Elbow) and silhouette score (Figure 1a). The silhouette is maximized at $k = 4$ ($s = 0.465$), which we adopt for the rest of the paper.

KMeans Objective. KMeans [10] minimizes

$$\mathcal{L}_{\text{KMeans}}(\{\mu_j\}) = \sum_{j=1}^k \sum_{x \in C_j} \|x - \mu_j\|_2^2, \quad (1)$$

where $\{\mu_j\}$ are cluster centroids and $\{C_j\}$ are the cluster assignments. The cluster identity assigned to each crash, $c_i \in \{0, \dots, 3\}$, is then exposed to the downstream classifier as a one-hot feature.

DBSCAN Comparison. As a non-parametric, density-based baseline [6], we run DBSCAN on a 25,000-crash sample of the training fold using the Haversine metric with $\varepsilon = 0.5$ km and `min_samples` = 50. DBSCAN identifies 31 clusters and labels 8,664 (35%) of the sample as noise, indicating that the underlying density field is highly heterogeneous and that a coarser KMeans-style partitioning is better suited to use as a categorical feature.

3.4 Severity Prediction Models

We approach severity prediction as binary classification and train a five-step ladder, all under the chronological split.

Majority Baseline. Always predict 0. This gives high accuracy (≈ 0.97) and zero recall on the serious class – a useful reference for how misleading raw accuracy can be under imbalance.

Logistic Regression. A class-weighted logistic regression

$$P(y = 1 \mid x) = \sigma(\beta_0 + \beta^\top x), \quad \mathcal{L} = - \sum_i w_{y_i} [y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad (2)$$

with $w_1/w_0 = N_0/N_1$ to handle imbalance.

Random Forest. A bagged ensemble of decision trees [3] with $n_{\text{est}}=300$, $\text{max_depth}=20$, and $\text{class_weight}=\text{balanced_subsample}$.

XGBoost. Gradient-boosted trees [5] with $n_{\text{est}}=400$, $\text{max_depth}=6$, $\text{learning_rate}=0.1$, and a positive-class weight $\text{scale_pos_weight} = N_0/N_1 \approx 36$. The optimisation objective is

$$\mathcal{L}_{\text{XGB}} = \sum_i \ell(y_i, \hat{f}(x_i)) + \sum_{m=1}^M \Omega(h_m), \quad (3)$$

where ℓ is binary log-loss and Ω regularizes tree complexity.

XGBoost + SMOTE. We fit the same XGBoost on training features re-sampled with the Synthetic Minority Over-sampling Technique [4].

XGBoost + Threshold Tuning. Holding the trained XGBoost fixed, we search the decision threshold τ that maximizes the F1 score of the serious class on the held-out validation slice, and evaluate at that threshold on the test set.

Feature-Group Ablation. For Logistic Regression, Random Forest, and XGBoost we run four feature sets: TIME-ONLY, +WEATHER, +WEATHER+SPATIAL, and +WEATHER+SPATIAL+CLUSTER (FULL). This isolates the marginal contribution of weather, the radial spatial feature, and the KMeans cluster identity.

4 Experiments & Results

4.1 Evaluation Metrics

We evaluate using accuracy, balanced accuracy, precision/recall/F1 for the serious class, ROC-AUC, and PR-AUC. Because the serious class accounts for only 2.75% of crashes, accuracy and ROC-AUC are insufficient on their own and we emphasize PR-AUC, balanced accuracy, and per-class recall.

4.2 Hotspot Discovery

Figure 1a reports the KMeans Elbow and Silhouette diagnostics: silhouette peaks at $k=4$ ($s = 0.465$). The resulting cluster map (Figure 2) recovers a recognizable Manhattan-South Bronx cluster (centroid 40.82, -73.90), a Brooklyn-South cluster (40.68, -73.95), an Eastern Queens cluster (40.70, -73.80), and a Staten Island cluster (40.60, -74.13). The serious-crash rate varies from 2.50% in the Brooklyn-South cluster to 3.21% in the Eastern Queens cluster – an approximately 28% relative gap that justifies the cluster id as a candidate feature (Figure 1b).

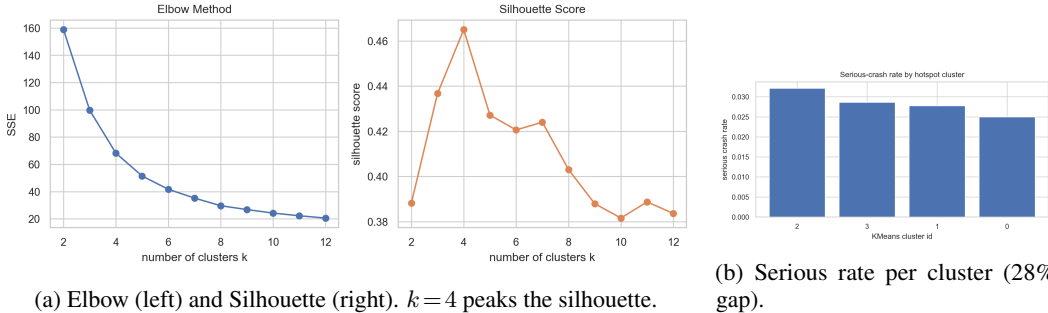


Figure 1: KMeans diagnostics and resulting cluster-level serious-crash rates.

Table 1 confirms that borough-level patterns broadly agree with the cluster-level analysis. Manhattan, despite being the densest crash area, has the *lowest* serious rate (1.51%); the Bronx, Brooklyn, and Staten Island sit between 2.5% and 2.6%. Table 2 shows the top 10 ZIP codes by serious-crash count – concentrated in Brooklyn and Queens (with one Bronx ZIP, 10469), consistent with the KMeans hotspots in Figure 2.

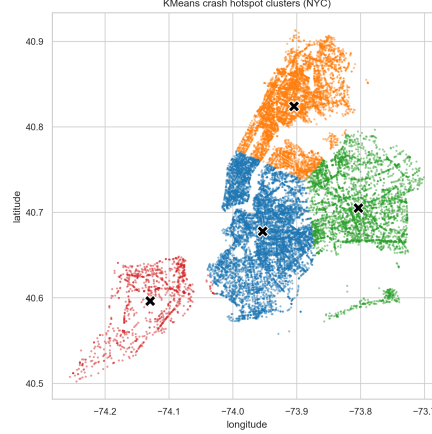


Figure 2: KMeans cluster assignment with centroids (black X). 20,000-crash random sample of the test fold.

Table 1: Borough-level totals and serious-crash rates (full 2021–2022).

Borough	Total	Serious	Serious %
Bronx	24,395	625	2.56
Brooklyn	46,564	1,183	2.54
Staten Island	5,111	129	2.52
Queens	36,376	855	2.35
Manhattan	22,240	336	1.51

4.3 Severity Prediction Results

Table 3 summarizes test-set performance for all models on the FULL feature set, plus the majority-class baseline. The headline findings are:

- **Accuracy is misleading.** The majority baseline already achieves 0.970 accuracy because the serious class is only 3% of the test set. Random Forest and XGBoost+SMOTE both collapse to near-majority predictions, recovering accuracy at the cost of zero serious recall.
- **Balanced accuracy and recall favor Logistic Regression.** The class-weighted logistic regression has the best balanced accuracy (0.585) and the best recall on the serious class (0.623). It does so at the cost of low precision (0.041), generating many false alarms – this is exactly the recall-precision tradeoff a deployed risk-flagging system has to negotiate.
- **PR-AUC is essentially flat across capacities.** Logistic Regression’s PR-AUC (0.045) is statistically indistinguishable from XGBoost’s (0.042) on this imbalanced task; Random Forest edges out XGBoost on ROC-AUC (0.593 vs. 0.588) but neither offers a meaningful gain over the linear baseline.
- **Threshold tuning trades recall for precision.** Selecting $\tau = 0.69$ on the validation slice (rather than the default 0.5) lowers XGBoost’s F1 on the serious class from 0.080 to 0.047 but raises precision from 0.048 to 0.056.
- **SMOTE hurts.** Synthetic minority oversampling collapses the model to majority predictions, consistent with prior reports that SMOTE on high-cardinality engineered features can blur decision boundaries [7].

Figure 3a shows the precision-recall and ROC curves for the three XGBoost variants. The PR curves remain flat and close to the random baseline, confirming the difficulty of the task; the ROC curves stay above the diagonal but only modestly. Figure 4 shows the corresponding confusion matrices: SMOTE collapses to majority predictions, default XGBoost flags many crashes as serious (high recall, many false positives), and the threshold-tuned XGBoost lands between the two with a much smaller false-positive count.

Table 2: Top 10 ZIP codes by serious-crash count (≥ 50 total crashes).

ZIP	Total	Serious	Serious %
11207	3,658	127	3.47
11236	2,367	99	4.18
11234	1,984	92	4.64
11212	2,358	73	3.10
11208	2,272	64	2.82
10469	1,264	64	5.06
11203	2,024	63	3.11
11434	1,534	52	3.39
11413	1,114	52	4.67
11226	1,774	45	2.54

Table 3: Severity-prediction test-set metrics (FULL feature set unless otherwise noted). PR-AUC and balanced accuracy are the headline metrics under imbalance.

Model	Acc.	Bal. Acc.	P (1)	R (1)	F1 (1)	PR-AUC	ROC-AUC
Majority baseline	0.970	0.500	0.000	0.000	0.000	—	—
Logistic Regression	0.550	0.585	0.041	0.623	0.077	0.045	0.615
Random Forest	0.969	0.501	0.115	0.003	0.005	0.044	0.593
XGBoost	0.837	0.544	0.048	0.231	0.080	0.042	0.588
XGBoost + SMOTE	0.969	0.500	0.091	0.001	0.002	0.040	0.583
XGBoost + Threshold ($\tau=0.69$)	0.950	0.510	0.056	0.041	0.047	0.042	0.588

4.4 Feature-Group Ablation

Table 4 reports XGBoost performance across the four feature sets. Adding weather features yields a small accuracy gain ($0.790 \rightarrow 0.839$) and a slight PR-AUC bump ($0.043 \rightarrow 0.044$); adding the radial spatial feature yields no additional benefit; adding cluster identity yields no benefit either, and in fact slightly hurts PR-AUC ($0.044 \rightarrow 0.042$).

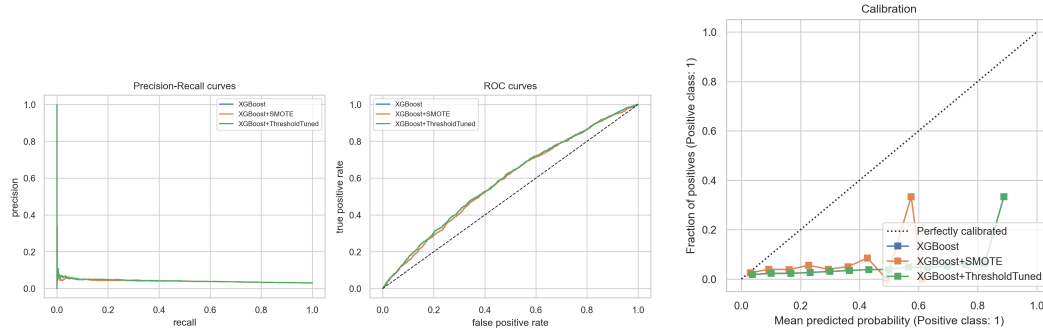
This is an interpretable negative result: gradient boosting can already extract spatial structure from the raw lat/lon coordinates, so a coarse 4-bin cluster identity is redundant. The cluster identity may still be valuable for downstream interpretability and for linear models like Logistic Regression (which gain a small bump in balanced accuracy when cluster id is added; see Tab. 3).

Table 4: XGBoost feature-group ablation. Weather contributes a small accuracy gain; spatial radial and cluster identity are redundant for tree-based models.

Feature set	Acc.	Bal. Acc.	PR-AUC	ROC-AUC	Recall (1)
time-only	0.790	0.557	0.043	0.598	0.309
+weather	0.839	0.555	0.044	0.588	0.252
+spatial	0.830	0.554	0.044	0.592	0.261
+cluster (full)	0.837	0.544	0.042	0.588	0.231

4.5 PCA and SHAP Interpretability

Figure 5 (left) projects the engineered numeric feature set onto its first two principal components: the serious class spreads across the projection rather than forming a separable cluster, which matches the modest PR-AUC numbers in Table 3. Figure 5 (right) reports the SHAP feature-importance bar plot for the FULL-feature XGBoost. Time and spatial features dominate (hour, sin / cos encodings, latitude, longitude, distance to midtown); weather features contribute meaningful but lower-ranked attributions (precipitation, cloud_pct, temp_max); cluster-id one-hot variables rank near the bottom, consistent with Table 4.



(a) PR (left) and ROC (right) curves for the three XGBoost variants.

(b) Calibration curves.

Figure 3: Test-set diagnostics for the XGBoost variants.

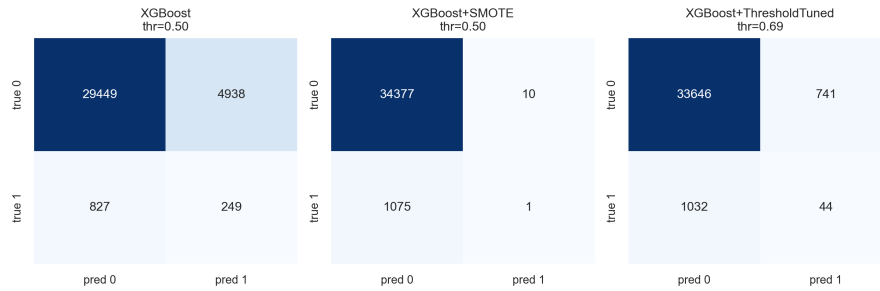
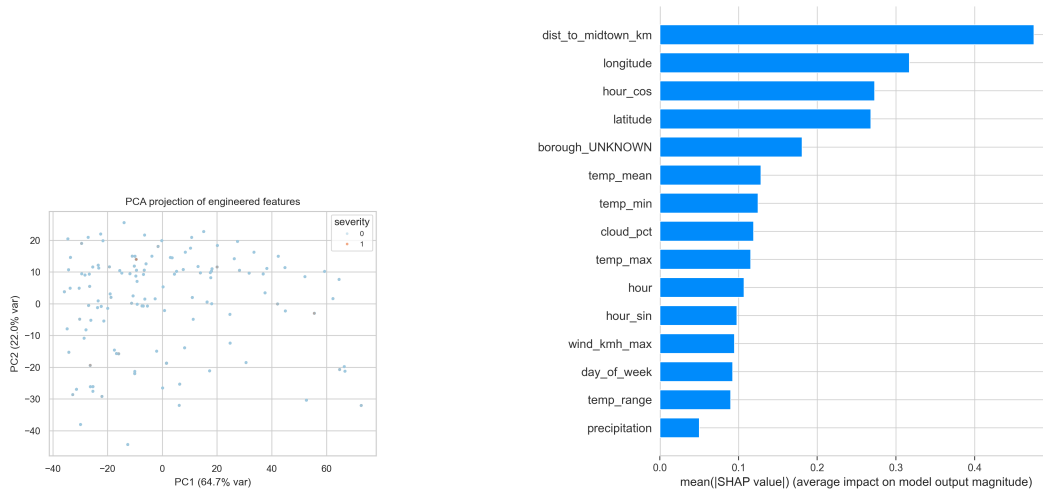


Figure 4: Confusion matrices for the three XGBoost variants. Among imbalance treatments, threshold tuning is the only one that avoids the SMOTE collapse to majority predictions while keeping the false-positive count manageable.



(a) PCA of engineered features (test set) by ground-truth severity.

(b) SHAP feature-importance bar plot, FULL-feature XGBoost.

Figure 5: Interpretability diagnostics for the trained XGBoost classifier.

5 Discussion and Conclusion

Three findings stand out from this hybrid pipeline. **Hotspot discovery is informative:** the KMeans clusters span an approximately 28% relative gap in serious-crash rate and are corroborated by the borough/ZIP analysis. **Recall is dominated by the imbalance treatment, not model capacity:** no model in the ladder reaches PR-AUC above 0.045, and a class-weighted Logistic Regression edges out the non-linear ensembles on balanced accuracy and PR-AUC – consistent with the imbalanced-learning literature [7] and attributable to the absence of causal features (vehicle speed/type, driver profile, road geometry). **Cluster identity is largely redundant for gradient boosting,** which already

extracts spatial structure from raw (lat, lon); the cluster id remains useful for linear models and for human-readable communication.

Limitations and future work. Dominant limitations are class imbalance with missing causal features, daily weather granularity, coordinate bias, and a two-year temporal scope. Promising next steps include hourly weather joins, NYC LION road features and DOT traffic counts [15], a learnable spatio-temporal grid, and a cost-sensitive evaluation metric for Vision Zero deployment.

Contributions

Pranav Arra (psa39): literature review and ablation design. Logan Sharrott (ls1428): unsupervised clustering and SHAP/PCA interpretability. Vishal Nagamalla (vn218): ETL pipeline, training ladder, and reproducibility. All three authors co-wrote the report.

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